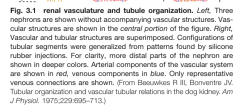


03. Renal Circulation and Glomerular Filtration · assets 1-9 / 27



Fig_3_1

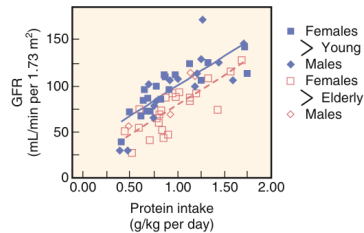


Fig. 3.4 Relationship between protein intake and glomerular filtration rate (GFR) determined from creatinine clearances. Data from younger (mean age, 31 years; range, 20–50 years) and older (mean age, 70 years; range, 55–88 years) healthy humans. *Closed symbols*, younger subjects; *open symbols*, older subjects; *squares*, women; *diamonds*, men. (Data from Lew SQ, Bosch P. Effect of diet on creatinine clearance in young and elderly healthy subjects and in patients with renal disease. *J Am Soc Nephrol*. 1991;2:856–865.)

Fig_3_4

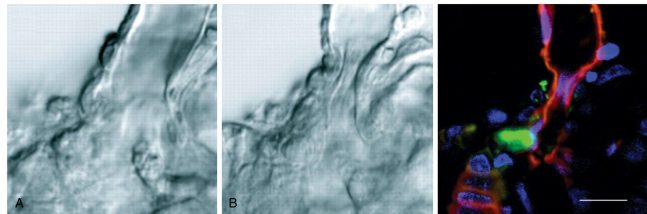
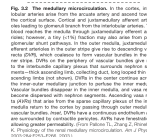


Fig. 3.7 Constriction of the terminal afferent arteriole (AA) via an intraglomerular precapillary sphincter in response to elevations in distal tubule NaCl content. (A, B) Transmitted light differential interference contrast (DIC) images. (A) Control, with NaCl concentration at the macula densa at 10 mM. (B) NaCl concentration is increased to 60 mM, resulting in an almost complete closure of the AA. (C) Fluorescence image of the same preparation as shown in B. Vascular endothelium and tubular epithelium are labeled with R18 (red), granules with quinacrine (green), and cell nuclei with Hoechst 33342 (blue). Note the presence of resin-positive granular cells in the AA sphincter segment and the highly contractile responses of the terminal portion of the AA. G, Glomerulus; MD, macula densa. Scale bar = 10 μ m. [From Pett-Peterson J. Multiphoton imaging of renal tissues in vitro. *Am J Physiol Renal Physiol*. 2005;288:F1079–F1083.]

Fig_3_7



Fig_3_2

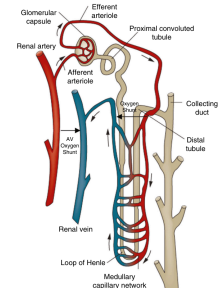


Fig. 3.5 Arterial to venous (A/V) oxygen shunts. Red blood cells in arterial vessels release oxygen, which diffuses into the interstitium to reach target cells. Blood with lower oxygen tension passes into the venous vessels. The juxtaposition of the arteries and veins present in the kidney facilitates oxygen diffusion from the artery to the vein. Thus oxygen tensions in the capillaries are relatively low when red blood cells reach the peritubular capillary plexus, particularly in the medullary capillary network and interstitium. (from Mmura I, Nangaku M. The suffocating kidney: tubulointerstitial hypoxia in end-stage renal disease. *Am J Kidney Dis* 2000;35:1091-1099)

Fig_3_5

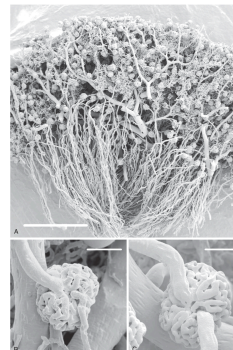


Fig. 18 (A) Resin cast of renal glomeruli of a rabbit. (B) Cortical glomerulus showing afferent (upper vessel) and efferent arterioles and the capillary tuft (scale bar = 60 μ m). (C) Juxtaglomerular glomerulus showing afferent (upper vessel) and efferent arterioles and the capillary tuft (scale bar 60 μ m). Note that the juxtaglomerular arterioles are larger in diameter than the cortical glomerular arterioles, particularly the efferent arterioles. (From Evans RG, Ezzamel WR, Denton KM. Mechanisms underlying the differential control of blood flow in the renal medulla and cortex. *J Hypertens*, 2004;22:1439-1451.)

Fig_3_8

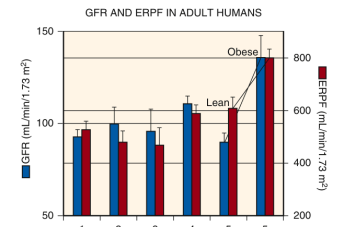


Fig. 3.3 Representative values for glomerular filtration rate (GFR) and estimated renal plasma flow (ERPF) from five studies in adults. Values from men and women were pooled. Numbers under each set of bars refer to the following studies: 1. Giordano and DeFronzo²⁶⁵; 2. Winet² et al.²⁶⁶; 3. Hostetter²⁶⁷; 4. Deen et al.²⁶⁸; 5. Chagnac and colleagues.²⁶⁹ For studies 1 through 3 and 5, values were obtained after approximately 1-hr periods of fasting; subjects in study 4 were allowed to eat. For study 5, values from lean subjects (average body mass index [BMI] = 22) were compared with those from obese nondiabetic individuals (BMI >38) after a 10-hr fast but are not corrected for body surface area.

Fig_3_3

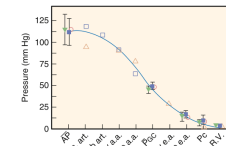


Fig. 3A Hydraulic pressure profile along the kidney vasculature. Filled squares and triangles denote values (mean $\pm$ 2 standard deviation) obtained from euolemic and hypotensive Munich rats. Values from studies in the squirrel monkey¹⁰ are shown as open circles. The pressure profile is similar in the two species. The pressure in the Sprague-Dawley rat in juxtamedullary nephrons followed with whole blood. These nephrons are located inside the cortical surface opposite to the pelvic incus and arcuate veins, in which entire pressure profiles can be obtained from the interlobular artery (arterio A), the interlobular vein (arterio V), the afferent arteriole (arterio A), the glomerular capillaries (P_G), the proximal (arterio e.a.) and late (arterio e.a.) segments of the efferent arteriole, the peritubular capillaries (P_{TC}) and the renal vein (R.V.). Arterial pressure: Arc; act, afferent arteriole; A, interlobular artery; V, interlobular vein; e.a., efferent arteriole; DA, Drenner's renal corpuscle; R.V., renal vein. *See text for details.* (From Brenner & The Kidney. 7th ed. Philadelphia: W.B. Saunders Company; 2004:353-412; Cassolas D, Navar LO. *In vitro* perfusion of juxtamedullary nephrons in rats. Am J Physiol. 1994;264:F534-F539; Inig J, Roman RL. Nephron pressure profile in glomerular capillaries. J Hypertens. 1992;10(Pt 2):770-774.)

Fig_3_6

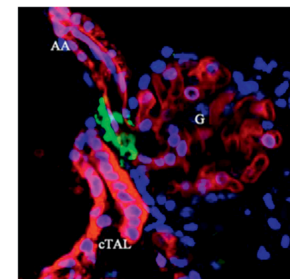


Fig. 3.9 Multicolor labeling of the *in vitro* microperfused juxtaglomerular apparatus with attached glomerulus. Cell membranes of tubular epithelium (cortical thick ascending limb, cTAL) containing the macula densa, vascular endothelium of the afferent arteriole (AA), and glomerulus (G) are labeled with R18 (red), renin granules with quinacrine (green), and cell nuclei with Hoechst 33342 (blue). (From Pet-Peterei J. Multispectral imaging of renal tissues *in vitro*. *Am J Physiol Renal Physiol*. 2005;288:F1079-F1083.)

Fig_3_9

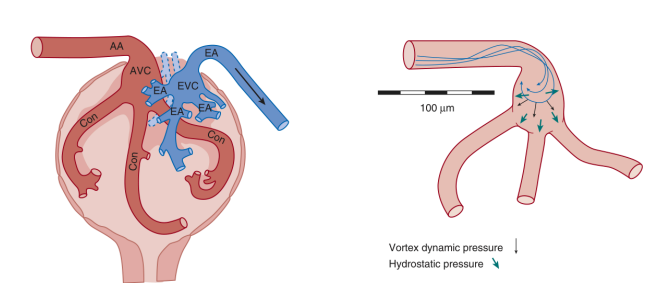


Fig. 3.10 Structure of the human glomerulus. Note the sharp turn made by the afferent arteriole (AA) as it enters the afferent vascular chamber (AVC), which flows into the conduit afferent capillaries (Con) and the efferent vascular chamber (EVC) transition into the efferent arteriole (EA). *Right*, Distribution of hydraulic forces in the AVC. (From Neal CR, Anil K, Ball JS, et al. Novel hemodynamic structures in the human glomerulus. *Am J Physiol Renal Physiol*. 2018;315(5):F1370–F1384; color figures courtesy Dr. Christopher Neal.)

Fig_3_10

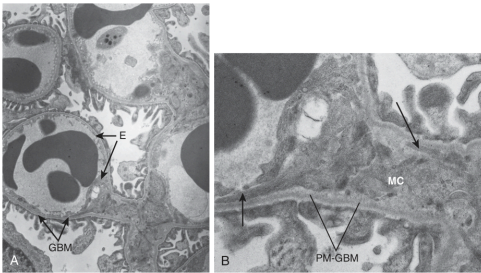


Fig. 3.11 Electron micrographs of glomerular capillaries of a Munich-Wistar rat. (A) An overview of several capillaries ($\times 14,500$). Most of the glomerular capillary endothelium (E) is in contact with the glomerular basement membrane (GBM) with only a small portion in contact with the mesangium (M). At its outer aspect, podocyte foot processes cover the GBM. There is no basement membrane separating the endothelium from the mesangium at their interface. (B) Mesangial cell (MC) extends outward to meet the glomerular capillary ($\times 42,000$). These cylinder-like stalks resemble contractile filament bundles (short arrow) that attach to the perimesangial glomerular basement membranes (PM-GBM) and extend to the GBM at the mesangial angles (long arrow).^{56,115,538} For this preparation, the nephron was perfusion fixed by micropuncture (D.A. Maddox) with 1.25% glutaraldehyde through the Bowman space.

Fig_3_11

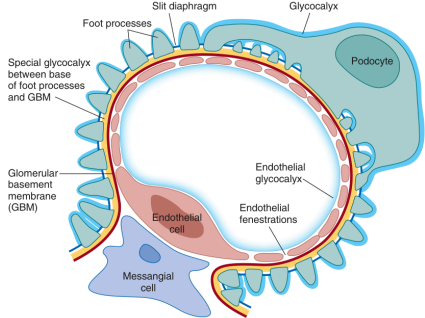


Fig. 3.12 Schematic drawing of a glomerular capillary. The elements considered as part of the glomerular filtration barrier include endothelial glycocalyx, fenestrated endothelium, basement membrane, specialized glycocalyx between the foot processes and basement membrane, and podocyte and slit diaphragm. (From Schlöndorff D, Wyatt CM, Campbell KN. Revisiting the determinants of glomerular filtration barrier: what goes round must come round. *Kidney Int*. 2017;92:533–536.)

Fig_3_12

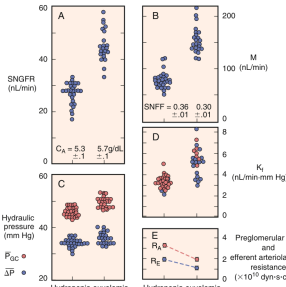


Fig. 3.13 Glomerular filtration in Munich-Wistar rats. (A–E) Each point represents the mean value reported for studies in hydropenic and euvoletic rats provided food and water ad lib until the time of study. Data from euvoletic rats are thought to be representative of nonanesthesia conditions. Only data from studies using male or a mix of male and female rats are shown. Values of the ultrafiltration coefficient, K_{uf} , and circles in D, denote minimum values because the animals were in filtration pressure equilibrium. Blue circles represent unique values of K_{uf} calculated under conditions of filtration pressure disequilibrium obtained primarily in euvoletic rats ($\Delta P \leq 0.95$). C_{cr} , Concentration in the afferent arteriole; ΔP , pressure gradient; P_{GC} , pressure in the glomerular capillaries; Q_{GC} , glomerular plasma flow rate; SNGFR, single-nephron filtration fraction; SNGFR, single-nephron glomerular filtration. (From Maddox DA, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The Kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412; Maddox DA, Deen WM, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The Kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412; Maddox DA, Deen WM, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The Kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412; Maddox DA, Deen WM, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The Kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412.)

Fig_3_13

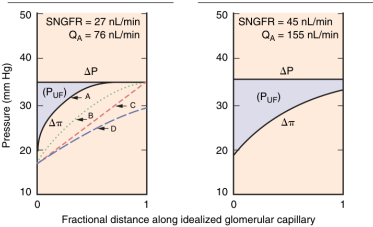


Fig. 3.14 Hydraulic and colloid osmotic pressure profiles along idealized glomerular capillaries in hydropenic and euvoletic rats. Values shown are mean values derived from the studies shown in Fig. 3.13. The transcapillary hydraulic pressure gradient, ΔP , is equal to $P_{GC} - P_T$, and the transcapillary colloid osmotic pressure gradient, $\Delta \pi$, is equal to $\pi_{GC} - \pi_{BS}$, where P_{GC} and P_{BS} are the hydraulic pressures in the glomerular capillary and the Bowman space, respectively, and π_{GC} and π_{BS} are the corresponding colloid osmotic pressures. Because the value of π_{BS} is negligible, $\Delta \pi$ essentially equals π_{GC} . P_{UF} is the ultrafiltration pressure at any point. The area between the ΔP and $\Delta \pi$ curves represents the net ultrafiltration pressure, P_{UF} . Left, Lines A and B represent two of the many possible profiles under conditions of filtration pressure equilibrium; line D represents disequilibrium and line C represents the linear $\Delta \pi$ profile. *Right*, Pressure profile after correction for surgical-induced loss of plasma volume depicts a small positive ΔP at the efferent level indicative of disequilibrium conditions. Q_{GC} , Glomerular plasma flow; SNGFR, single-nephron glomerular filtration rate.

Fig_3_14

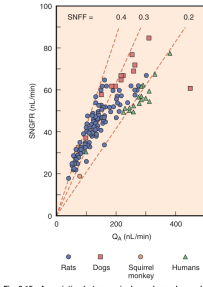


Fig. 3.15 Association between single-nephron glomerular filtration rate (SNGFR) and glomerular plasma flow (Q_{GC}). Data are from rats, dogs, squirrel monkeys, and humans. The values for SNGFR and Q_{GC} for humans were calculated by dividing the whole-kidney GFR and renal plasma flow by the estimated total number of nephrons/kidney (1 million). Each point represents the mean value for a given study. SNGFR, Single-nephron filtration fraction. (Data from Maddox DA, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The Kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412; Maddox DA, Deen WM, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The Kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412; Maddox DA, Deen WM, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The Kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412.)

Fig_3_15

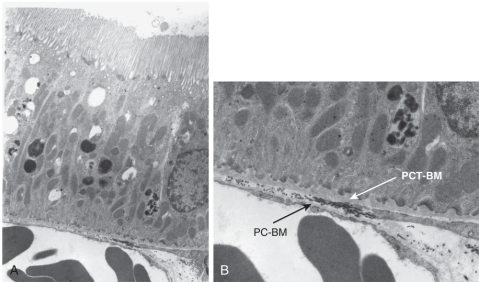


Fig. 3.16 Electron micrographs (by D.A. Maddox) of a proximal tubule of a Munich-Wistar rat. Tubule wall was perfusion fixed with 1.25% glutaraldehyde, thereby also fixing red cells in adjacent capillaries. (A) The apposition of the basolateral surface of the tubular cells with the adjacent peritubular capillaries is close, leaving little interstitial space where the two come in contact. ($\times 13,000$). (B) The proximal tubule basement membrane (PCT-BM) is relatively thick in comparison with the peritubular capillary endothelial basement membrane (PC-BM). ($\times 25,000$.)

Fig_3_16

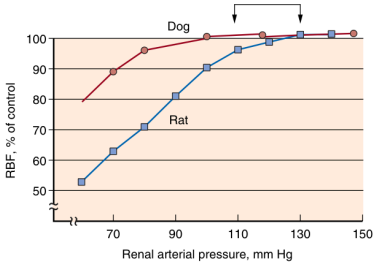


Fig. 3.17 Autoregulatory response of total renal blood flow (RBF) to changes in renal perfusion pressure in the dog and rat. In general, the normal anesthetized dog exhibits greater autoregulatory capability to maintain RBF and glomerular filtration rate to lower arterial pressures than the rat. (From Navar LG, Ball PD, Burke TJ. Role of a macula densa feedback mechanism as a mediator of renal autoregulation. *Kidney Int*. 1982;22:157–164.)

Fig_3_17

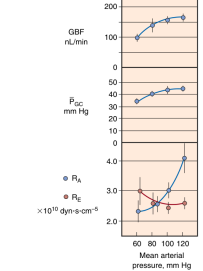


Fig. 3.18 Glomerular dynamics in response to reductions of renal arterial pressure in the normal hydropenic rat. As can be seen, glomerular blood flow (GBF) and glomerular capillary hydraulic pressure (P_{GC}) remain relatively constant as blood pressure is lowered from ~ 120 to ~ 80 mm Hg over the range of perfusion pressure examined, primarily because of reductions in afferent arteriole resistance (R_{AF}). Efferent arteriole resistance (R_{EF}) was relatively constant but increased slightly at lower pressures. (Modified from Robertson CR, Deen WM, Toy JL, Brenner BM. Dynamics of glomerular ultrafiltration in the rat. II. Hemodynamics and autoregulation. *Am J Physiol*. 1972;223:1191, 1972.)

Fig_3_18

Fig_3_19

Fig_3_22

Fig_3_25

Fig. 3.20 Schematic of representative single-nephron glomerular filtration (SNGFR) rate responses to changes in flow to the distal nephron (TGF) and drawing showing the micropuncture procedure used to change the delivery of fluid to the distal nephron (courtesy Pamela Carmine). The SNGFR decreases with increases in distal volume delivery due to a TGF-mediated increase in afferent arteriolar constriction and subsequent decrease in glomerular pressure with an increase in responsiveness at flows in the normal operating range. TGF responsiveness is dynamic and is modulated by changes in the interstitial environment. The interstitial environment is altered by changes in fluid volume or levels of synthesis of renin-angiotensin system components, administration of vasodilators or vasoconstrictors, and administration of Ang II, Angiotensin II; HETE, hydroxyoctadecatrienoic acid; NOS, nitric oxide; NOS, nitric oxide synthase; PGJ₂, prostaglandin I₂.

Fig_3_20

Fig. 3.23 (A–F) Role of nitric oxide in the control of glomerular filtration dynamics. Studies were performed in euolemic Munich-Wistar rats receiving intravenous pressor doses of the nonselective nitric oxide synthase (NOS) blocker, N-monomethyl-L-arginine (NMA; i.v., filled squares), or nonpressor doses of NMA at the origin of the renal artery (i.r., open squares). K_f , filtration coefficient; P_{DO} , pressure in the afferent arteriole; P_T , pressure in the tubules; O_{GFR} , glomerular plasma flow rate; R_{PG} , preglomerular arterial resistance; R_{PE} , efferent arterial resistance; $CapGFR$, single-nephron glomerular filtration rate. [Data (mean $\pm$ SE) obtained from Deng & Baylis C. Locally produced NO controls preglomerular

Fig_3_23

Fig. 3.26 Correlation of aging with systolic blood pressure and creatinine clearance (C_{Cr}) in normal adults. (From Lindeman R, Tobin J, Shock N. Association between blood pressure and the rate of decline in renal function with age. *Kidney Int.* 1984;26: 861–868.)

Fig_3_26

[illegible]

Fig_3_21

Fig. 3.24 (A–D) Effects of intravenous administration of endothelin (subpressor dose) on glomerular dynamics. K_f Filtration coefficient; ΔP , mean transcapillary pressure gradient; P_{GO} , mean glomerular capillary pressure; Q_{NG} , single-nephron glomerular plasma flow; R_{eff} , afferent arterial resistance; R_{eff} , efferent arterial resistance; SN_{GFR} , single-nephron glomerular filtration rate. [Data [mean $\pm$ SEM] obtained in Munich-Water rats from Back KF, et al. Mesangial cell, glomerular and renal vascular responses to endothelin in the rat kidney. Elucidation of signal transduction pathways. *J Clin Invest*. 1989;83(1):336–342.]

Fig_3_24

Table_3_1